HiPAD: A Unified Gait, Audio-Visual, and Personality Fusion Network for Depression Detection, with a Hierarchical Personality Ablation and the Calibration Limits of Domain-Adapted LLMs

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Abstract

We present **HiPAD**, a single fusion network submitted to all six sub-tracks of the MPDD-AVG 2026 depression-detection challenge, covering the Elder and Young cohorts and three modality sets—gait, audio-visual, and both—each combined with personality descriptions. The network encodes each available modality with a shared BiLSTM, zero-pads absent modalities, and applies cross-modal attention only when at least two of gait, audio, and video are present. Personality, which dominates our results, is kept out of the attention and fused by concatenation. Because the architecture is identical in every sub-track, differences between sub-tracks reflect the data rather than the model, and we use this to study three questions. Our main finding is that the personality description—roughly 150 tokens of text about who the subject is—carries more depression signal than audio, video, and gait combined: gait plus personality alone (TLS 0.647) beats every configuration that includes audio and video. First, a four-level ablation of the personality description (L0–L3) quantifies this: removing every personality-derived feature, including the Big Five scores that earlier MPDD ablations keep, lowers the Track-Level Score by 0.499 to 0.507 and leaves the model at chance. Second, domain-adapting the personality encoder with LoRA (7B) or QLoRA (72B) on PHQ-8/9 data raises binary Cohen's κ by 0.13 but lowers the overall score by distorting the PHQ-9 regression; a frozen RoBERTa-large encoder does better at this sample size. Third, at fixed architecture, handcrafted clinical features (eGeMAPS, OpenFace action units) outperform generic deep features (wav2vec2, DenseNet) on the audio-visual tracks. The unified system reaches a cross-track G-Score of 0.482; with fewer than 100 subjects per cohort, the limiting factors are the size of the representation and the calibration of the outputs, not the choice of architecture.

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CCS Concepts

• **Computing methodologies** → **Machine learning**; *Neural networks*; • **Applied computing** → *Health informatics*.

Keywords

depression detection, multimodal fusion, personality modelling, large language models, calibration, audio-visual, gait, BiLSTM, PHQ-9, ablation study

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1 Introduction

Depression affects over 280 million people globally [34]. Automated assessment benefits from multiple behavioural modalities: personality traits capture stable dispositional risk factors [17], while audio-visual cues and gait patterns encode momentary psychomotor symptoms. The MPDD-AVG 2026 challenge [8, 9] organises six sub-tracks across Elder (≥60 years) and Young cohorts, combining gait (IMU), audio-visual interviews, and—uniquely—free-text *personality descriptions* encoding Big Five traits, demographics, and disease history.

Challenge entries are typically specialised per sub-track, which precludes cross-track conclusions; and prior personality-aware MPDD systems (P3HF [10], Fosafer [30]) measure the personality modality against *soft*-ablation baselines that retain numerical Big Five scores. We address both gaps with one system evaluated on all six sub-tracks, and make four contributions:

(1) **A unified variable-modality fusion network.** HiPAD encodes whichever modalities a sub-track provides with a shared BiLSTM design, activates cross-modal attention only between ≥2 non-personality modalities, and fuses the dominant personality stream by concatenation only (Section 4.1); one capacity-matched configuration (h=128) covers five of six sub-tracks unchanged, reaching a cross-track G-Score of 0.482.

(2) **A true hierarchical personality ablation.** A four-level decomposition (L0–L3, Section 4.2) whose L0 baseline removes *all* personality-derived features shows the personality-description modality contributes +0.499–0.507 average Track-Level Score, and that L0 models are at chance (Section 5.2).
(3) **The calibration limit of domain-adapted LLMs.** LoRA 7B and QLoRA 72B personality encoders raise binary κ by +0.13 but lower the overall score through PHQ-9 regression calibration mismatch, isolated with hybrid task-head experiments (Section 6.1).
(4) **Cohort-dependent feature selection under one architecture.** Clinically-motivated features (eGeMAPS [7], OpenFace [3] action units) beat generic deep features in every same-architecture audio-visual comparison (Section 5.4), and one capacity-vs-n analysis explains why larger hidden widths, larger embeddings, and extra modalities all fail to help at n=87–88 (Section 6.3).

2 Related Work

Depression detection has been studied using audio features [4, 5, 7], video features [3], text transcripts [38], audio-visual shared tasks [27, 29], and gait [32]; see Song et al. [28] for a recent survey. Semi-structured interviews remain the dominant elicitation protocol [24], the PHQ-9 questionnaire [18] the standard severity measure, and BiLSTMs with attention pooling remain competitive on small sequential behavioural datasets [19]. The MPDD dataset [8, 9] uniquely adds free-text personality descriptions as an explicit modality—relevant because high Neuroticism and low Conscientiousness are among the strongest Big Five predictors of depression risk [17]—but prior MPDD systems (P3HF [10], Fosafer [30]) measure the personality contribution against *soft*-ablation baselines that retain the numerical Big Five scores. On the encoder side, LLMs perform strongly on clinical text [37], and LoRA [14] enables domain adaptation of 7B–72B models on small clinical datasets such as DAIC-WOZ (51 subjects) [12]; however, fine-tuning commonly degrades LLM calibration even when raw task performance improves [35, 40], and DAIC-WOZ adaptation risks learning disorder-general distress cues rather than depression-specific markers [25]—both directly relevant to the failure mode we isolate in Section 6.1. Among the audio-visual features themselves, eGeMAPS [7] provides 65 handcrafted features with established depression biomarkers [33] and OpenFace [3] action units map onto depressive symptoms [11], a clinical specificity that generic representations such as wav2vec2 [2] and DenseNet [15] lack. An extended review is given in the supplementary material.

3 Task and Dataset

3.1 Task Formulation

Per subject, the challenge provides (by sub-track) gait IMU sequences, audio-visual interview recordings, and a personality description $\mathbf{d}$ (English text, ≈150 tokens) with structural features $\mathbf{s}$ (Big Five scores, age, sex, financial stress, family size, and disease class for Elder). Interview audio (Mandarin) is transcribed with Whisper large-v3 [26] and encoded with Chinese-RoBERTa [6] (768-dim [CLS]), mean-pooled over a subject's 3–4 sessions into a transcript embedding $\mathbf{t}$. We predict **binary** status (depressed: PHQ-9 $\geq$ 10), **ternary** severity (none $<$ 5 / mild 5–10 / severe $\geq$ 10), and continuous **PHQ-9** $\hat{r} \in [0, 27]$.

Following the official scoring rules [8, 9], the Track-Level Score is TLS = (Cls_F1 + CCC + κ)/3, where Cls_F1 (macro-F1) and κ each average their binary and ternary counterparts and CCC is the PHQ-9 concordance correlation coefficient [20]. Tables additionally report *ternary* Cohen's κ as a supplementary diagnostic, being more sensitive to chance-level agreement on imbalanced severity classes. The cross-track *G-Score* is the mean TLS over all six sub-tracks.

3.2 Dataset and Sub-Tracks

Three modality sets—G+P (gait), A-V+P (audio-visual), A-V-G+P (both), each with personality—are paired with the Elder and Young cohorts. **Elder**: 87 train, 23 test subjects; descriptions include a disease class (neurological, endocrine, circulatory, other, healthy), clinically relevant for late-life depression. **Young**: 88 train, 22 test subjects; no disease class. The cohorts' training PHQ-9 distributions have similar means (Young 4.8 vs. Elder 4.5) and severe counts (10 vs. 11), but Young is *compressed*: SD 3.5 vs. 4.7 with nearly twice as many mild-band subjects (33 vs. 17), concentrated near the ternary thresholds (Section 6.4).

Transcript note. The transcript embedding $\mathbf{t}$ enters the personality stream in *all* sub-tracks, including G+P; the sub-track definitions describe intended modality focus rather than an eligibility rule, and the organisers' reproducibility verification reran our released pipeline, transcript included. The headline personality gain is not a transcript artefact: the transcript is present on *both* sides of the ablation, and L0 (gait+transcript, no personality) is at chance (Section 5.2).

4 The HiPAD Architecture

4.1 Unified Variable-Modality Design

HiPAD (*Hierarchical Personality-Aware Depression* model) is instantiated for each sub-track only through its modality list (Section 3.2, Figure 1; feature matrix in the supplementary material); in the released code it is `EnhancedMPDDModel`. Each available non-personality modality $m \in \{g, a, v\}$ (gait, audio, video) is encoded by a 2-layer bidirectional LSTM with temporal attention pooling and layer normalisation, $\mathbf{z}_m = \mathrm{LN}\big(\sum_t \alpha_t^{(m)}\, \mathrm{BiLSTM}_2(\mathbf{f}_m)_t\big)$ with $\alpha^{(m)} = \mathrm{softmax}\big(W_a\, \mathrm{BiLSTM}_2(\mathbf{f}_m)\big)$, where $\mathbf{f}_m \in \mathbb{R}^{T_m \times d_m}$ is the raw frame sequence (d_g gait, d_a∈{65, 768} audio, d_v=710 video), linearly projected to the model width h=128. h is the *final bidirectional* output width (each direction of the 2-layer LSTM has hidden size $h/2$=64), so each pooled $\mathbf{z}_m \in \mathbb{R}^h$ and the fused width below is $h \cdot |\mathcal{M}|$; the temporal attention $W_a \in \mathbb{R}^{1\times h}$ assigns one scalar weight per time step (softmax over t). Audio and video arrive as up to 4 independently-recorded interview-segment "pairs" per subject (zero-padded with a binary mask when fewer are available); each segment is encoded by the *same* BiLSTM and combined across segments by a second learned softmax pooling layer (*LearnedPairPool*). The personality stream concatenates the personality embedding $\mathbf{e}_p$ with small-MLP projections of the 18-dim structural features $\mathbf{s}$ and the transcript $\mathbf{t}$, then passes through a 3-layer MLP with a residual

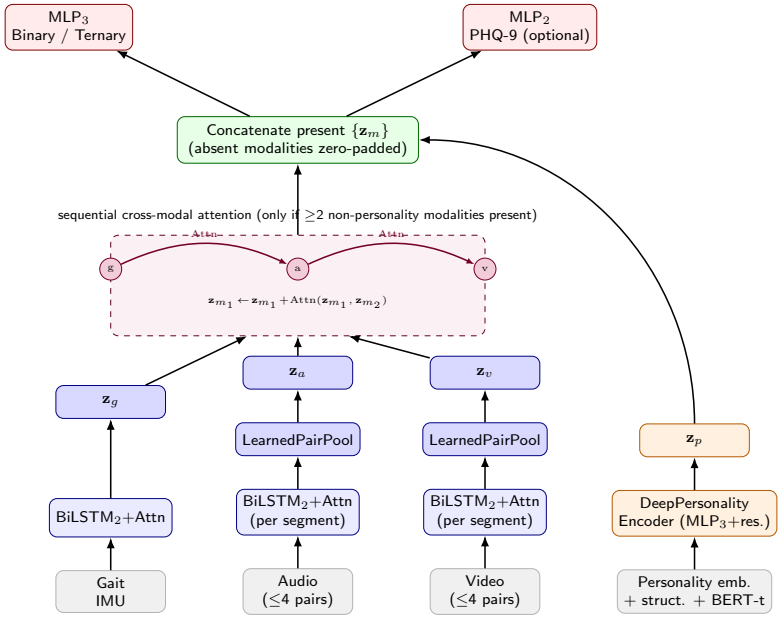


Figure 1: HiPAD pipeline. Gait, audio, and video pass through cross-modal attention *only when at least two are present*; personality always bypasses attention and joins at concatenation. Audio and video arrive as up to 4 interview segments, pooled by a learned softmax layer; absent modalities (per sub-track, Section 3.2) are zero-padded.

shortcut (*DeepPersonalityEncoder*):

$$\mathbf{z}_p = \mathrm{LN}\big(\mathrm{MLP}_3(\mathbf{x}) + W_{\mathrm{res}}\mathbf{x}\big), \quad \mathbf{x} = [\mathbf{e}_p;\, \mathrm{MLP}_s(\mathbf{s});\, \mathrm{MLP}_t(\mathbf{t})], \quad (1)$$

with $\mathbf{z}_p \in \mathbb{R}^h$ and W_{res} a linear map from the concatenated input to $\mathbb{R}^h$.

When two or more non-personality modalities are present (A-V+P, A-V-G+P), each ordered pair (m_1, m_2) updates its representation by residual single-head scaled dot-product attention, $\mathbf{z}_{m_1} \leftarrow \mathbf{z}_{m_1} + W_o\,\mathrm{softmax}\big((W_q\mathbf{z}_{m_1})(W_k\mathbf{z}_{m_2})^\top/\sqrt{h}\big)\,W_v\mathbf{z}_{m_2}$, with $W_q, W_k, W_v, W_o \in \mathbb{R}^{h\times h}$. Pairs are visited in the fixed order gait, audio, video and applied sequentially; we did not tune this order. For G+P, with a single non-personality modality, no cross-modal attention is applied. Personality is fused only by concatenation, never by cross-attention: in this small-n regime, letting the dominant personality signal attend to the much weaker A-V streams degraded validation performance. The per-modality vectors are concatenated (fused_dim = $h \times |\mathcal{M}|$) and passed to a 3-layer MLP classification head (binary or ternary, trained as separate models per task and ensembled) and, when enabled, a parallel 2-layer MLP regression head (PHQ-9).

Features. Audio: OpenSMILE eGeMAPS v2 (65-dim) or wav2vec2-base (768-dim), per Section 5.4. Video: OpenFace 2.0 (710-dim). Gait: IMU sequences with stride, cadence, and asymmetry features. Personality: RoBERTa-large [22] (1024-dim [CLS]) plus 18-dim structural features (Big Five [16], demographics, Elder disease class), present in all sub-tracks. Full feature-extraction settings are in the supporting materials.

4.2 Personality Encoding Hierarchy

We evaluate four levels of representation of the challenge's *personality-description modality*, whose free text bundles Big Five traits with demographics and, for Elder, disease history. We measure this modality's contribution as a whole; "personality gain" throughout means *personality-description-modality* gain, without claiming demographics or disease class are personality traits—disease class in particular is a direct clinical correlate of late-life depression [13].

- **L0 — No personality (true ablation).** Only the gait encoder and transcript remain; all personality-derived features are removed: no text embedding, no Big Five structural features, no demographic or disease flags.
- **L1 — Structural features only.** The description parsed into the 18-dim vector $\mathbf{s}$: Big Five (N, E, A, O, C $\in [1, 15]$), age, sex, financial stress (0–4), family members, Elder disease class (one-hot). No text embedding.
- **L2 — Description embeddings.** Frozen RoBERTa-large (1024-dim [CLS]) plus $\mathbf{s}$.
- **L3 — Domain-adapted LLM.** `gte-Qwen2-7B-instruct` [1] fine-tuned via LoRA (r=64, α=128) in two stages—PHQ-8 regression on DAIC-WOZ [12], then PHQ-9 regression on the MPDD train set (3584-dim); `Qwen2.5-72B-Instruct` [36] via 4-bit QLoRA (r=64), same pipeline (8192-dim); and `e5-mistral-7b-instruct` [31] (4096-dim, frozen) as a non-adapted control.

4.3 Training

Each task head is trained separately. Binary and ternary heads use focal loss [21] (γ=2.0, label smoothing 0.1); the ternary head adds an ordinal expected-rank penalty (λ_{ord}=0.5); the auxiliary PHQ-9 head, when enabled, adds an MSE term (λ_{reg}=0.3) on $\log(1+\cdot)$-normalised scores. Optimisation: AdamW, batch size 8, up to 120 epochs, early stopping (patience 30 on validation macro-F1), weight decay 10^{-5}

Table 1: Track-Level Scores at challenge close (2026-06-01); Ter $\kappa^{\ddagger}$ supplementary diagnostic, Rank† interim standing (final ranking: Section 8). The italicised Elder A-V+P row§ is from a separate, non-HiPAD pipeline (Section 5.1), excluded from all cross-track comparisons; the unified G-Score uses the HiPAD value (0.397, only the aggregate TLS available).

Sub-track	TLS	Bin κ	Ter $\kappa^{\ddagger}$	CCC	Rank†
Elder G+P (16079)	**0.647**	0.572	0.677	0.549	#7/31
Young G+P (16035)	**0.578**	0.636	0.405	0.527	#2
Elder A-V+P (16078)	0.397	—	—	—	—
Elder A-V-G+P (16077)	0.477	0.330	0.319	0.504	#14/47
Young A-V-G+P (16034)	0.397	0.455	0.103	0.338	#2
Young A-V+P (16033)	0.396	0.364	0.272	0.300	#2
Elder A-V+P, non-HiPAD§	*0.484*	0.439	0.434	0.374	#10/34
G-Score (unified HiPAD)	**0.482**				
G-Score (best submitted§)	0.496				

(G+P) / 10^{-4} (A-V), MixUp [39] (α=0.2, p=0.5), learning rate 8×10^{-5} (G+P) / 3×10^{-4} (A-V). We use k-fold cross-validation (k=4 main, k=8 for the ensemble-depth study) with 6 model seeds per fold (all seeds listed in the supporting materials), ensembling the top-5 checkpoints by validation κ. Features are z-score-normalised per feature on training-split statistics. Ternary decoding (G+P): Elder uses the ternary head's argmax; Young thresholds the model's own regression output $\hat{r}$ at 5.0/7.0 (thresholds on predictions, not labels). Hardware details are given in the supporting materials.

Reproducibility. Retrainable checkpoints and configurations for five of the six sub-track configurations, the training script `train10.py`, and a one-command inference script for the final G+P configurations are bundled in the released evaluation package; the encoder ablations (L0, L1, L3) are retrainable from the released code and per-encoder embeddings. The supporting materials give the full feature-extraction and optimisation settings, the CodaBench submission IDs for every number reported here, the experiment-ID-to-checkpoint mapping, and the per-run logs.

5 Results

5.1 Cross-Track Scores

G+P sub-tracks dominate (avg TLS 0.612 vs. 0.417 for A-V), reflecting the decisive contribution of personality descriptions, quantified next; on the organisers' reproducibility-verified final leaderboard, our entries place 2nd on all three Young sub-tracks (Section 8).

The Elder A-V+P entry. Our best-submitted Elder A-V+P predictions (TLS 0.484) came from a separate pipeline developed independently by the second author (`ReliabilityGatedEnhancedModel`: text-reliability gating with personality-conditioned cross-attention), without k-fold cross-validation and released only as precomputed predictions. It is evidence neither about HiPAD nor about audio features, so we exclude it from every cross-track comparison and headline claim; it appears only in the separated rows of Tables 1 and 4. The retrainable HiPAD configuration (eGeMAPS+OpenFace) reaches 0.397 here; disentangling the +0.087 gap would require retraining both pipelines under a common protocol, which the released artefacts do not permit (Section 8).

Table 2: Personality-description-modality gain on the official CodaBench leaderboard. L0: gait+transcript only, *every* personality-derived feature removed; L2: +18-dim structured features and RoBERTa-large embedding; "final": best task-specific ensemble.

Level	Features	Elder	Young	ΔE	ΔY
L0	Gait+transcript, no personality	0.099	0.113	—	—
L2	+struct + RoBERTa-large (k=4)	0.632	**0.578**	+0.533	+0.465
final	+task-specific ensemble	**0.647**	**0.578**	+0.548	+0.465

5.2 Hierarchical Personality Contribution

We measure the personality-description-modality gain on the official CodaBench leaderboard (Table 2) and isolate the *structured-vs-textual* split on held-out validation (full table in the supplementary material), because the leaderboard closed before a structured-only configuration could be scored on the test set. All conditions use the identical architecture, k=4 folds, 6 seeds.

The modality lifts TLS by **+0.499** on average at the protocol-matched L2 level and **+0.507** with the final ensembles; it also dwarfs every A-V score in Table 1. This gain is an order of magnitude larger than prior MPDD reports (P3HF +0.055 accuracy [10], Fosafer +0.029–0.057 F1 [30]; metrics not directly comparable, table in the supplementary material), for two structural reasons: soft ablations retain the Big Five scores in the baseline, and in G+P gait alone is a weak baseline, mechanically enlarging the *relative* contribution. L0 models collapse to chance (binary κ = 0.00, both cohorts): gait+transcript alone carries no depression-severity signal in this architecture, so the entire gain is attributable to the personality-description modality (we diagnose the near-zero L0 itself in Section 6.2).

Structured Big Five and demographic features alone (L1) already recover strong classification on validation (Elder binary κ=0.67, Young 0.53); adding the RoBERTa-large description embedding (L2) improves Young markedly (binary κ 0.53→0.81, ternary 0.41→0.75), while for Elder the two are comparable on validation—an instance of the validation-to-leaderboard gap analysed in Section 6.3. We report these as relative validation comparisons because validation overestimates held-out performance in this small-n regime.

5.3 Personality Encoder Comparison

Table 3 compares personality encoders across the hierarchy on the official leaderboard.

5.4 Audio Feature Comparison

eGeMAPS wins every same-architecture comparison (Table 4), and during development DenseNet video features likewise underperformed OpenFace action units and were not submitted. The best Elder A-V+P entry nevertheless used wav2vec2 (0.484), but it is the non-HiPAD pipeline of Section 5.1, so its margin says nothing about the audio representation; whether self-supervised features capture elderly phonation cues that eGeMAPS misses [23] remains untested (extended discussion in the supplementary material).

Table 3: Encoder comparison on official CodaBench LB scores. †val max κ=0.91 vs. LB 0.439 (overfitting; Section 6.3). "final": binary k=8 + ternary/PHQ k=4 ensembles.

Encoder	Dim	Bin κ	Ter $\kappa^{\ddagger}$	CCC	**TLS**
Elder G+P (16079)					
None (L0)	—	0.000	0.082	0.030	0.099
RoBERTa-large (L2)	1024	0.646	**0.677**	0.549	0.632
LoRA 7B (L3a)	3584	**0.776**	0.553	0.415	0.582
L3a$_{bin}$+BERT$_{ter}$ (L3b)	hyb	**0.776**	**0.677**	0.420	0.624
72B$_{QL}$+BERT$_{ter}$ (L3c)	hyb	0.735	**0.677**	0.480	0.631
QLoRA 72B full (L3d)	8192	0.735	0.480	0.358	0.543
e5-mistral	4096	0.439†	0.311	0.443	0.473
Bin-8 + Tern/PHQ-4 (final)	hyb	0.572	**0.677**	**0.549**	**0.647**
Young G+P (16035)					
None (L0)	—	0.000	0.000	0.006	0.113
RoBERTa-large (L2)	1024	0.636	0.405	0.527	**0.578**
LoRA 7B (L3a)	3584	0.564	0.177	0.395	0.411

Table 4: Audio feature choice vs. A-V Track-Level Score at fixed HiPAD architecture; the non-HiPAD Elder A-V+P entry (Section 5.1) below the rule is not a same-architecture comparison.

Sub-track	Cohort	wav2vec2	eGeMAPS	Winner
A-V+P (16078)	Elder	—	0.397	—
A-V-G+P (16077)	Elder	0.459	**0.477**	eGeMAPS
A-V+P (16033)	Young	0.336	**0.396**	eGeMAPS
A-V-G+P (16034)	Young	0.386	**0.397**	eGeMAPS
A-V+P (16078), non-HiPAD§	Elder	*0.484*	—	—

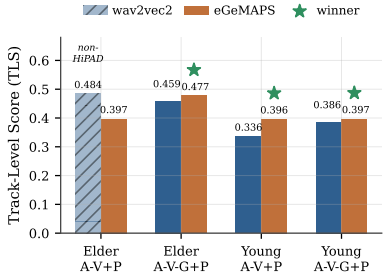


Figure 2: wav2vec2 vs. eGeMAPS Track-Level Score across the A-V sub-tracks (stars mark winners); the Elder A-V+P wav2vec2 bar is the non-HiPAD entry (Section 5.1).

6 Analysis

6.1 LLM Calibration Mismatch

Table 3 reveals a dissociation (visualised in the supplementary material): LoRA 7B reaches binary κ = 0.776 (+0.13 over RoBERTa-large) yet a *lower* TLS (0.582 vs. 0.647). TLS weights F1, κ, and CCC equally, so degraded regression or ternary quality directly cancels binary gains. The mechanism is calibration: LoRA 7B fine-tuned on PHQ regression produces extreme continuous predictions (validation range 0.67–9.78 vs. 2.0–5.0 for RoBERTa-large), which (1) collapses CCC (0.549→0.415) and (2) misfires the fixed ternary thresholds (5.0/7.0) applied to out-of-range outputs (ternary κ 0.677→0.553). The hybrid L3b configuration (LoRA binary head + BERT ternary head) confirms the diagnosis: restoring ternary quality raises TLS from 0.582 to 0.624 with binary unchanged. Domain-adapted LLMs therefore require post-hoc PHQ-9 calibration (isotonic regression or Platt scaling) before deployment in TLS-optimised systems. This extends the finding that fine-tuning degrades LLM calibration [35, 40] to a concrete metric drop in clinical regression, compounded by the DAIC-WOZ→MPDD domain gap [25].

6.2 Diagnosing the Chance-Level L0 Baseline

Both cohorts' L0 models collapse to majority-class prediction (binary κ = 0.00), and two readings must be separated. As an *ablation baseline*, L0 shows the personality-description modality—not gait—is the primary discriminative signal in G+P; since a baseline retaining Big Five scores already reaches binary $\kappa \approx 0.53$–0.67 on validation (Section 5.2), soft designs substantially understate the modality's contribution. As a statement *about gait*, L0 is only a lower bound for this architecture class—gait detection on larger cohorts reports meaningful discriminative power [32], and the challenge provides no external no-personality baseline here—so we claim only that a capacity-matched BiLSTM at $n \approx 90$ recovers no depression-severity signal from IMU gait plus transcript.

6.3 Capacity Must Match n=87–88

Every change that added capacity failed to transfer to the test set, whereas deeper ensembling, which adds no parameters, helped. **Hidden width**: h=256 matched or slightly exceeded h=128 on Elder validation (κ 0.732 vs. ~0.72), yet all four submitted h=256 variants regressed on the leaderboard (−0.004 to −0.038 TLS); Young regressed on both. **Embedding dimensionality**: e5-mistral (4096-dim) reached validation binary κ = 0.91 but the lowest submitted TLS (0.473, Table 3)—at n=87, a 4096-dim input memorises validation splits, and unlike the regression-specific LoRA failure, this degrades all three task heads. **Ensemble depth** is the complementary lever, adding no parameters: raising k from 4 to 8 gained +0.021 TLS on Young, and a task-specific hybrid combining the k=8 binary ensemble with the k=4 ternary/PHQ ensemble gained +0.015 on Elder (0.632→0.647); the hybrid, however, cherry-picks checkpoints across two k-fold runs, which makes it sensitive to reproduction variance—plausibly why it is absent from the organisers' independently reproduced ranking (Section 8). Full ensemble depths per configuration are in the supporting materials. We therefore use h=128 and ≤1024-dim embeddings uniformly, which also enables cross-track analysis without confounding architecture effects. The same signature explains why *adding modalities can hurt*: the A-V-G+P tracks score below gait-only G+P (Elder 0.647→0.477; Young 0.578→0.397) despite being a modality superset, because concatenating two weak, noisy A-V streams beside a dominant personality signal enlarges the fused width and raises capacity without matching signal—diluting the decisive modality exactly as extra hidden units do, and answering the apparent paradox in Table 1 (visualised as a radar chart in the supplementary material).

6.4 Cohort Effects: Elder vs. Young

Elder outperforms Young on every same-architecture comparison: G+P +0.069 TLS, A-V-G+P +0.080, while A-V+P is effectively tied

(0.397 vs. 0.396, Table 4). The evidence localises the gap in severity grading, not detection: Young ternary κ is far below Elder's under identical models (A-V+P: 0.272 vs. 0.434; G+P: 0.405 vs. 0.677), whereas Young *binary* κ is the best we measured on G+P (0.636). This matches the label distributions (Section 3.2): Young's PHQ-9 scores are compressed (SD 3.5 vs. 4.7) with twice the mass in the mild band, i.e., concentrated near the ternary thresholds, so equal-quality predictions yield more boundary errors. It is also consistent with—though not direct evidence for—older adults presenting more somatic and psychomotor symptoms, younger adults more cognitive ones [13]. The same asymmetry recurs inside the personality modality (Elder's gain +0.548 vs. Young's +0.465, partly because Elder descriptions include disease class, a direct clinical correlate) and in gait: holding HiPAD/eGeMAPS fixed, adding gait to A-V+P gains +0.080 TLS for Elder but +0.001 for Young (Table 4)—again the psychomotor-channel pattern; and domain-adapted LLMs hurt Young more (ternary κ 0.405→0.177), consistent with the DAIC-WOZ fine-tuning distribution diverging further from Young's compressed PHQ profile. Together this suggests Young-cohort progress needs richer linguistic features or finer-grained severity annotation, not further architecture changes.

7 Discussion

The personality ablation and the system results describe the same effect from two sides: on the gait tracks, L0 shows that personality is essentially the only informative modality, and on the audio-visual tracks the same imbalance explains why adding modalities can lower the score (Section 6.3).

Three of our observations extend prior work rather than repeat it: the personality gain of 0.499 to 0.507 shows that the small soft-ablation gains reported by P3HF and Fosafer are mostly an artefact of leaving the Big Five scores in the baseline (Section 5.2); the calibration failure of the adapted LLMs is a clinical-regression instance of fine-tuning degrading LLM calibration [35, 40], with the hybrid-head experiment pinning the damage to the regression output; and eGeMAPS and OpenFace beating generic deep features at $n<100$ carries the AVEC-era experience with handcrafted paralinguistic features [27, 33] over to a new dataset and cohort split.

For practitioners with small clinical cohorts: a frozen mid-sized encoder with deeper ensembling is safer than a domain-adapted LLM unless the regression output is re-calibrated after adaptation (e.g., isotonic regression or Platt scaling), and a result from a different model class should not be used to draw conclusions about features (Section 5.4). Promising directions are reliability-weighted fusion tolerating missing modalities, personality-conditioned attention, and adaptation to non-Chinese populations. On the evaluation side, per-component score reporting (F1, CCC, κ) and a challenge-defined true-ablation protocol would make calibration failures visible and personality-gain numbers comparable across teams.

8 Limitations

Each cohort provides only n=87–88 training and 22–23 test subjects: this forces h=128, makes cross-validation noisy, and lets a handful of misclassified subjects shift leaderboard positions; we therefore report no confidence intervals or significance tests, and controlled module ablations (cross-modal attention, LearnedPairPool, residual personality encoder, single-model vs. ensemble) remain future work. The unified-architecture claim holds for five of six sub-tracks; the sixth is the non-HiPAD Elder A-V+P entry (Section 5.1), released only as precomputed predictions rather than retrainable weights.

Final ranking. The interim CodaBench standings at challenge close were superseded by the organisers' reproducibility-verified final leaderboard, which we treat as authoritative: it confirms rank 2nd on all three Young sub-tracks; our Elder entries do not appear in its published top-3, so Elder scores in Table 1 are our CodaBench scores at challenge close. All submissions were finalised before the deadline and are reproducible from the released checkpoints, configurations, and logs.

The clinical population is exclusively Chinese (Mandarin interviews, English personality descriptions), so generalisability to other languages and populations is untested; and only the aggregate TLS was exposed per submission, limiting post-hoc error analysis.

Ethical use. This system detects statistical correlates of depression from personality, speech, face, and gait—sensitive health data—and is intended only as *screening support* to prioritise clinical attention, never as standalone diagnosis; deployment would require consent, clinician oversight, bias auditing across demographic groups, and validation beyond the single cohort studied here.

9 Conclusion

We submitted one fixed BiLSTM fusion network to all six MPDD-AVG 2026 sub-tracks, comparing cohorts and modality sets without architecture confounds. The one result a reader should take away is how much the personality description matters: 150 tokens of text about who the subject is predict depression better than audio, video, and gait combined, and a model with gait and personality alone (TLS 0.647) beats every configuration that adds audio and video; a true ablation puts the contribution at 0.499–0.507 TLS, and without it the model is at chance. Domain-adapting the personality encoder improves classification but distorts the PHQ-9 regression, so a frozen RoBERTa-large encoder with deeper ensembling is the better choice at this scale; handcrafted audio-visual features beat generic deep features in every fixed-architecture comparison; and larger hidden widths, larger embeddings, and extra modalities all overfit alike on fewer than 100 subjects. The main obstacles at this sample size are representation size and output calibration rather than architecture—PHQ-9 calibration in particular deserves more attention in multi-task depression assessment.

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